Gaussian Process Regression: Kernel Inductive Bias, Posterior Uncertainty, and Bayesian Optimisation

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***Abstract*—This report examines Gaussian process regression across five experiments, starting from the construction of GP priors to Bayesian optimisation. Four kernel families are classed through their covariance matrices and prior sample paths, and the qualitative signatures that allow or exclude specific families from consideration given observed data are identified. Three kernels are fitted to the regression target $f(x) = 1.5x + \sin(x)$: the squared-exponential achieves $\log \mathrm{ML} = -11.32$, the Ornstein-Uhlenbeck -21.11, and the linear kernel -22.87. Leave-One-Out (LOO) analysis on the SE model yields a training RMSE of 0.195 and a LOO RMSE of 0.272 that depicts a slight bias in the marginal likelihood objective. An over-smoothed GP with $\ell = 15$ is shown to produce confidently incorrect extrapolations despite smooth posterior means. Under expected improvement, the calibrated surrogate identifies $x = 6.5$ as the next query with $\alpha_{\mathrm{EI}} = 1.024$; we examine the limitations and the effect of noise level on the exploration-exploitation balance alongside this result.**

***Index Terms*—Gaussian processes, Bayesian optimisation, kernel methods, surrogate modelling, inductive bias.**

I. Introduction

A Gaussian Process (GP) is a tool that defines a prior distribution over functions, where providing an input results in a full distribution reflecting uncertainty in the predicted output, whereby statistical nature is determined by the covariance kernel $k(x_m, x_n)$ [1] being used. The kernel makes use of prior beliefs regarding several factors; (i) smoothness, (ii) stationarity, and (iii) periodicity before observing any data, these form the inductive bias of the model. Two GPs trained on identical data with varying kernel choice can match everywhere the data sits and split completely as they extrapolate out of bounds. The choice of kernel being the primary reason for their discrepant behaviours.

In Bayesian Optimisation (BO), a GP surrogate guides the search for a function's maximum via an acquisition function. Miscalibrated kernels collapse the acquisition to almost zero (in an over-confident scenario, explored in Section IV). This may direct the "query budget" to regions that do not offer much promise, often times missing unexplored peaks.

Section II configures different kernels to examine their behavious as covariance matrices and prior sample paths vary, identifying inferrable properties from finite samples. Section III fits three kernels to a regression benchmark to quantify inductive bias through marginal likelihood. Section IV analyses posterior variance behaviour, model assessment via lead-one-out (LOO) cross-validation, as well as overconfidence. Section V develops the expected improvement acquisition function and constructs a failure scenario while we explore the role of noise in optimisation.

II. Gaussian Process Priors and Covariance Structure

A. Two-dimensional covariance and correlation

Before examining GP priors over function spaces, we observe covariance at the level of a single 2×2 matrix. Defining

$$K = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}, \quad -1 < c < 1, \tag{1}$$

$N = 50$ samples are drawn from $\mathcal{N}(\mathbf{y}; \mathbf{0}, K)$, each of the form $(y_{n1}, y_{n2})^\top$, and plotted as line segments connecting the coordinates at indices 1 and 2. Fig. 1 shows the result for $c \in \{-0.9, -0.5, 0, 0.5, 0.9\}$.

At $c = 0.9$, the components are strongly positively correlated. We observe y_{n2} tending to have the same sign and magnitude as y_{n1}, judging from their more horizontal, uniformly directed lines. The correlation inverts at $c = -0.9$; lines appear mostly negatively sloped. The transition through to $c = 0$ is most symmetric. Independence between y_{n1} and y_{n2} in this particular case reveals slope direction is randomly determined. It is clear that the off-diagonal entry of a covariance matrix affects the strength at which two pairs of random variables co-vary.

B. Covariance kernels for GP priors

With $N = 100$ uniformly spaced inputs $x_1 < \cdots < x_N$ on $[0, 5]$, the GP prior covariance matrix has elements $K_{mn} = k(x_m, x_n) + \sigma_d^2 \delta_{mn}$, where $\sigma_d = 0.1$ is the noise standard deviation and δ_{mn} is the Kronecker delta. Four kernels are implemented with fixed parameters $\sigma_f = 1.0$ and $\ell = 1.0$.

The squared-exponential (SE) kernel,

$$k_{\mathrm{SE}}(x_m, x_n) = \sigma_f^2 \exp\left(-\frac{\|x_m - x_n\|^2}{2\ell^2}\right), \tag{2}$$

produces an infinitely differentiable covariance. Over distance, this covariance decays quadratically. The Ornstein-Uhlenbeck (OU) kernel,

$$k_{\mathrm{OU}}(x_m, x_n) = \sigma_f^2 \exp\left(-\frac{\|x_m - x_n\|}{\ell}\right), \tag{3}$$

decays linearly in distance and corresponds to the Matérn-1/2 family, producing continuous but non-differentiable sample

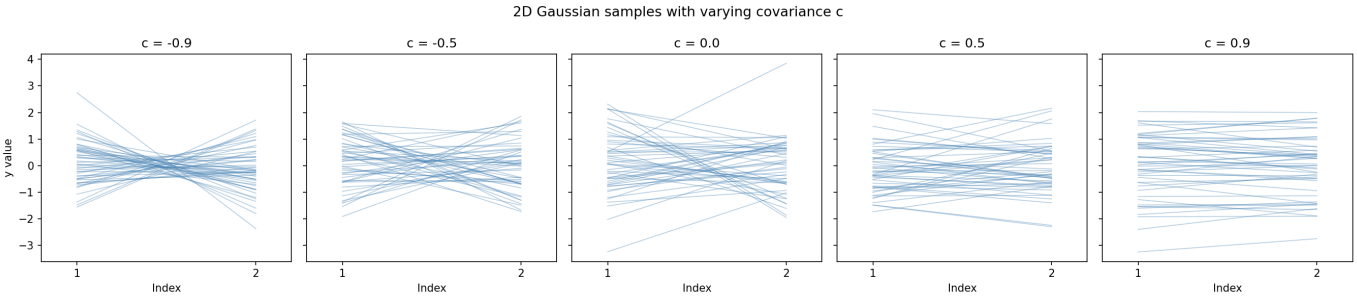


Fig. 1: Line plots of $N = 50$ samples from $\mathcal{N}(\mathbf{0}, K)$ for five values of c.

paths. The oscillatory kernel combines a periodic "suppression term" with a Gaussian envelope:

$$k_{\text{osc}}(x_m, x_n) = \sigma_f^2 \exp\left[-(A \sin \tfrac{\pi}{P} d)^2 - \frac{d^2}{2\ell^2}\right], \quad (4)$$

where $d = \|x_m - x_n\|$, $A = 1.0$, and $P = 2.0$. The linear kernel,

$$k_{\text{lin}}(x_m, x_n) = \sigma_f^2 (x_m - c)(x_n - c), \quad (5)$$

with $c = 2.5$ is the only non-stationary kernel in the set. Covariance depends on absolute input positions.

The four 100×100 covariance matrices, in Fig. 2 reveal the following. The SE band is wide and smooth, diagonally-oriented. It reflects long-range quadratic decay of correlation because the squared term in (2) indicates that the covariance between x_m and x_n decays in accordance to the square of their distance. OU on the other hand is narrower due to faster exponential decay at shorter distances, as introduced by the exponential term in (3). The OSC matrix modulates the banding pattern with a periodic ripple at the characteristic period $2P = 4$. Finally, the linear matrix displays no banding, structured instead as an outer product with high positive covariance in the inputs coming from $c = 2.5$ and negative covariance everywhere else.

C. Prior sample paths

Samples from $\mathcal{N}(\mathbf{0}, K)$ are drawn by Cholesky decomposition. If $K = LL^\top$, then $\mathbf{y} = L\mathbf{z}$ where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$. Five samples per kernel are shown in Fig. 3.

The sample paths for the SE configuration are smooth all throughout. No kinks or abrupt direction changes. OU paths are continuous everywhere but non-differentiable, sharp angular turns appear at every scale uniformly across the domain. In the case of OSC paths, we see periodic oscillation at the characteristic period $2P = 4$ units, with varied amplitude across samples albeit a consistent frequency structure. Linear paths are perfectly straight lines that cross zero at $x = 2.5$, proceeding by diverging linearly. That zero-crossing anchor is a feature of the non-stationary prior; prior variance is zero at the pivot c following construction, regardless of where else the function goes.

D. Inferring kernel families from sample paths

The task is to identify plausible from excludable kernel families given a set of observed sample paths. The principal diagnostic properties are smoothness of sample paths, stationarity, and periodicity.

For SE samples, two plausible families are the SE kernel itself and the Matérn-5/2 kernel. Both produce smooth, infinitely or twice-differentiable paths that are visually indistinguishable over five samples on a domain of length 5. To distinguish them, we examine how the variance of finite differences $\text{Var}(f(x+h)-f(x))$ scales as $h \to 0$; with a grid of 100 points and five samples, that test has negligible statistical allure. We rule out OU paths for SE samples since they are non-differentiable and display sharp angular kinks that are absent from SE paths.

For OU samples, Matérn-3/2 is a plausible alternative; both produce rough, non-smooth paths. The difference in their small-scale increment scaling exponents cannot be resolved from a small sample. Conversely, SE is excluded: no choice of length scale produces the sharp, uniformly distributed kinks visible in Fig. 3.

For linear samples, we exclude all stationary kernels, as they produce samples with uniform statistical character across the domain. The linear samples show very small variance approaching $x = 2.5$ and growing variance further away. Hyperparameter tuning with the use of stationary kernels would not be able to reproduce this property.

III. Inductive Bias Through Kernel Choice

A. Dataset and motivation

The regression target is $f(x) = 1.5x + \sin(x)$ with $N = 15$ uniformly spaced points on $[0, 5]$ and additive Gaussian noise $\sigma_{\text{noise}} = 0.3$. We choose this function because of its dominant linear trend making the linear kernel a good prior: $f(0) = 0$ places the pivot at $c = 0$, and the linear component accounts for the majority of the function's range over $[0, 5]$. Furthermore, the sinusoidal oscillation at period $2\pi \approx 6.3$ is slow enough for 15 uniformly spaced points to resolve it at an SE length scale of $\ell \approx 2.75$. The test domain extends to $[-1.5, 6.5]$, with a margin of $\delta = 1.5$ beyond each training bound.

B. Hyperparameter optimisation

Hyperparameters for all three kernels are estimated by maximising the log marginal likelihood:

$$\log p(\mathbf{t}|X, \boldsymbol{\theta}) = -\tfrac{1}{2}\mathbf{t}^\top K^{-1}\mathbf{t} - \tfrac{1}{2}\log|K| - \tfrac{N}{2}\log 2\pi, \quad (6)$$

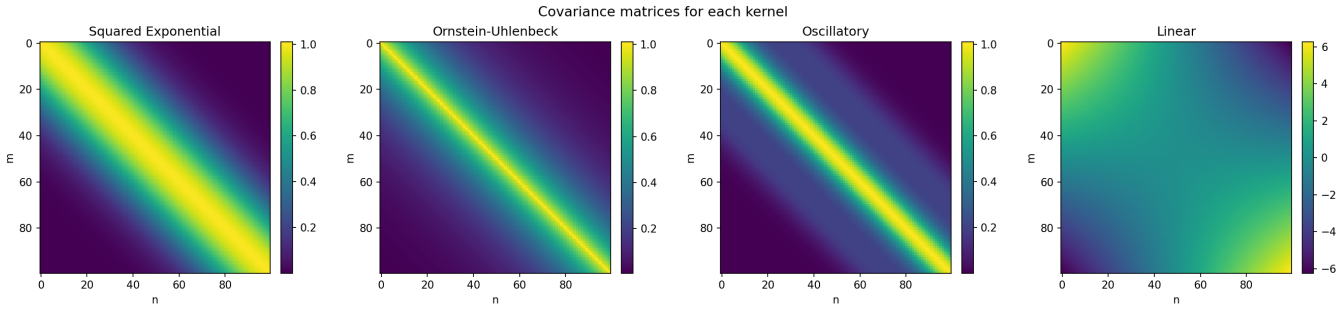


Fig. 2: Covariance matrices for the four kernel families on $N = 100$ inputs over $[0, 5]$, $\sigma_f = 1$, $\ell = 1$, $\sigma_d = 0.1$. From left to right: SE, OU, oscillatory, linear ($c = 2.5$).

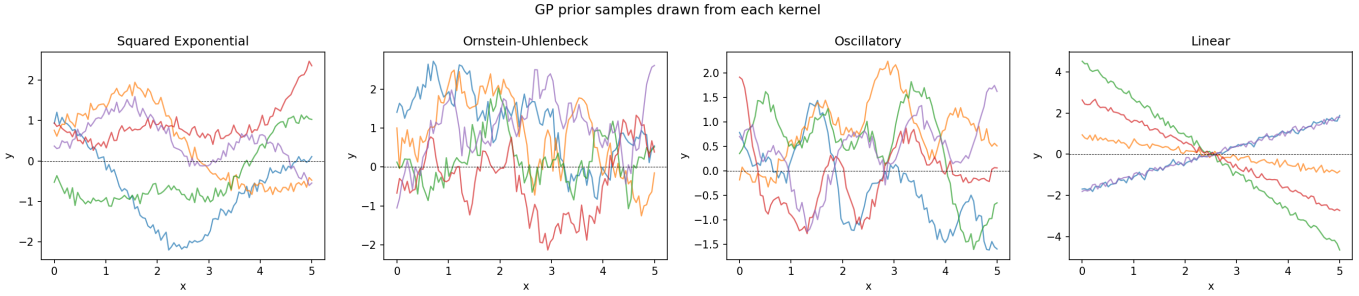


Fig. 3: Five prior sample paths for each of the four kernels over $[0, 5]$. From left to right: SE (smooth, infinitely differentiable), OU (continuous but non-differentiable), oscillatory (periodic at scale $2P = 4$), linear (exactly linear through $c = 2.5$).

where $K = k(X, X) + \sigma_d^2 I$ [1]. Optimisation uses L-BFGS-B in log parameter space with three random restarts to mitigate local optima. For the OU kernel, the length scale upper bound is constrained to half the domain length ($\ell \leq 2.5$) to prevent a "degenerate" solution in which all training points become correlated and the noise drops toward zero. The linear kernel's pivot is fixed at $c = 0$.

Optimised hyperparameters and marginal likelihoods are given in Table I.

TABLE I: Optimised hyperparameters and log marginal likelihoods for the three fitted GP models.

Kernel	$\hat{\sigma}_f$	$\hat{\ell}$	$\hat{\sigma}_d$	log ML
SE	4.816	2.753	0.232	-11.32
OU	1.888	2.500	0.100	-21.11
Linear	1.398	—	0.919	-22.87

C. Posterior comparison

The three posteriors are shown in Fig. 4. All three achieve reasonable fits within the training range; the differences are prominently depicted in the extrapolation region.

Within $[0, 5]$, the SE posterior has the narrowest confidence interval. With $\hat{\ell} = 2.753$, the kernel resolves both the linear trend and the sinusoidal variation, and the posterior mean tracks the training data closely. The OU posterior is wider due to the roughness prior being violated by the smooth target, and the optimiser compensating by choosing the maximum allowed $\hat{\ell} = 2.500$. The linear posterior is narrow near $x = 0$ (where prior variance is zero) and expands gradually towards $x = 5$.

The extrapolation behaviour is where the kernels diverge mostly. The SE posterior mean reverts towards zero on both sides of $[0, 5]$ with what seems to be symmetrical growing uncertainty; as the distance from all training inputs grows, the SE kernel correlation decays to zero and the posterior converges to the prior. The OU model behaves similarly. The linear posterior mean extends the fitted line indefinitely in both directions. Uncertainty on the left side remains small because those inputs are near the anchor $c = 0$; on the right side the prior variance grows as $\hat{\sigma}_f^2 x^2$, and the Confidence Interval (CI) grows outward, likely due to non-stationarity.

D. Inductive bias analysis

The marginal likelihood ordering SE $>$ OU $>$ Linear quantifies the degree of prior mismatch with the data. The SE kernel places a prior over infinitely differentiable functions varying at scale ℓ; the correct prior for the smooth target achieving the best logML by approximately ten log-units over the nearest kernel.

The OU kernel places a roughness prior: functions are only mean-square continuous, with sharp local direction changes. The true function violates this. The optimiser's response, choosing the maximum allowed $\hat{\ell} = 2.500$, effectively forces the OU kernel to behave more like a slowly varying smooth process. The logML of -21.11 reflects complexity cost of a prior that does not match the data's differentiability structure.

The linear kernel represents the strongest prior where the model class is restricted entirely to lines through $x = 0$.

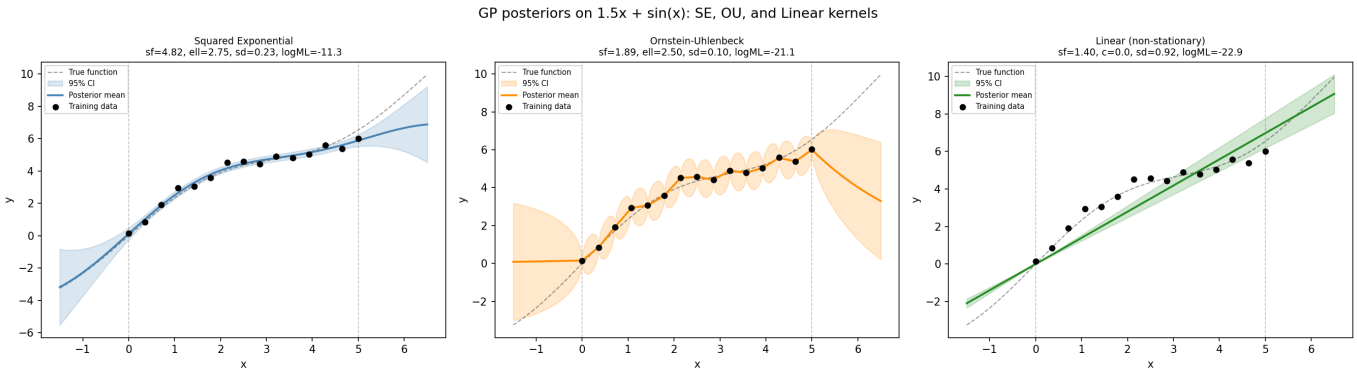


Fig. 4: GP posteriors on $f(x) = 1.5x + \sin(x)$ for the SE (left), OU (centre), and linear (right) kernels. Each panel shows the true function (dashed), posterior mean, 95% CI, and training data. Vertical dashed lines mark the training boundaries at $x = 0$ and $x = 5$.

This prior is agreeable for when domain knowledge supports linearity (like when the function is known to pass through a fixed point). The linear model absorbs the sinusoidal residual as noise, increasing $\hat{\sigma}_d = 0.919$ against the true noise of 0.3. The logML of -22.87 shows this property is expensive.

Inductive bias is not inherently a flaw. If the data were generated by a rough process or a truly linear function, the logML ordering would invert. The marginal likelihood provides a good comparison because it integrates over the fitted parameters and quantifies how well each prior accounts for observed data as a whole, not needing a held-out validation set [1].

IV. Posterior Uncertainty and Confidence Intervals

A. Posterior formulas and implementation

The GP posterior for M test inputs X^* is characterised by the mean and covariance:

$$\mu_{t^*} = k(X^*, X)\big(k(X, X) + \sigma_d^2 I\big)^{-1}\mathbf{t}, \tag{7}$$

$$\Sigma_{t^*} = k(X^*, X^*) - k(X^*, X)\big(k(X, X) + \sigma_d^2 I\big)^{-1}k(X, X^*). \tag{8}$$

Both are computed via the Cholesky factorisation of $K = k(X, X) + \sigma_d^2 I$, avoiding explicit matrix inversion and being more numerically stable [1]. The 95% confidence interval at each test point is $\mu(x^*) \pm 1.96\,\sigma(x^*)$, where $\sigma(x^*) = \sqrt{\max(\mathrm{diag}(\Sigma_{t^*}), 0)}$.

B. Posterior variance behaviour

The SE posterior is shown in Fig. 5. At training inputs, the posterior variance is not zero. The model treats observations as noisy ($\sigma_d^2 > 0$), so uncertainty cannot be reduced below the noise minimum. The posterior at a training location is about $\sigma_d^2 = 0.054$. An interpolating GP ($\sigma_d = 0$) would collapse the variance to zero at each training point and pass exactly through every observation.

Between training inputs, variance rises above the nearest training-point values. With $\hat{\ell} = 2.753$ and grid spacing 0.357, the kernel correlation between consecutive training inputs is

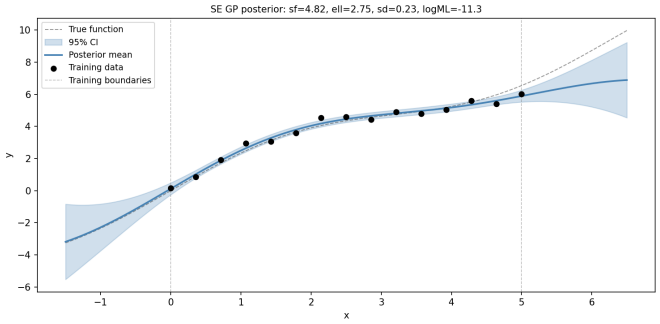


Fig. 5: SE GP posterior with MLE hyperparameters $\hat{\sigma}_f = 4.816$, $\hat{\ell} = 2.753$, $\hat{\sigma}_d = 0.232$. The shaded band is the 95% CI.

$\exp\!\big(-0.357^2/(2 \times 2.753^2)\big) \approx 0.991$. Adjacent points are almost fully correlated, it follows that the posterior remains tightly constrained throughout $[0, 5]$ and the CI is narrow across the whole training range.

At points farther from observed data, the variance grows monotonically toward the prior variance $\sigma_f^2 = 23.2$. As the distance from all training inputs increases, the SE kernel correlation decays to zero. The posterior receives no information from the training set and collapses onto the prior, producing the fanning CI that extends past both boundaries in Fig. 5.

C. Marginal likelihood trend and leave-one-out error

The log marginal likelihood as a function of ℓ (with fixed $\hat{\sigma}_f$ and $\hat{\sigma}_d$) is plotted in Fig. 6. The curve is unimodal with peak $\hat{\ell} = 2.753$.

For $\ell < 1$, the logML falls steeply. Short length scales allow the GP to explain each training point with an isolated spike, reducing the data-fit term $-\frac{1}{2}\mathbf{t}^\top K^{-1}\mathbf{t}$ but incurring a large log-determinant penalty $-\frac{1}{2}\log|K|$. For $\ell > 3$, the logML decreases more gradually because the kernel over-smooths the sinusoidal variation, and the data-fit term deteriorates. The fact that the curve is asymmetric ties to the notion that overfit models are penalised more sharply than underfit ones for this dataset. The marginal likelihood peak follows a form of

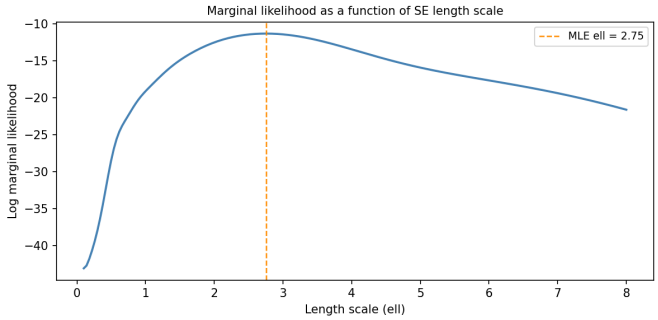


Fig. 6: Log marginal likelihood as a function of SE length scale ℓ, with $\hat{\sigma}_f = 4.816$ and $\hat{\sigma}_d = 0.232$ held fixed. The peak at $\hat{\ell} = 2.753$ (orange dashed line) is the maximum likelihood length scale.

Bayesian Occam's Razor, being that complexity is penalised automatically without requiring a validation set [1].

Leave-one-out (LOO) cross-validation provides another insight. Using the identities [1]:

$$\sigma^2_{\mathrm{LOO},i} = 1/[K^{-1}]_{ii}, \tag{9}$$

$$\mu_{\mathrm{LOO},i} = t_i - [K^{-1}\mathbf{t}]_i/[K^{-1}]_{ii}, \tag{10}$$

the full set of LOO predictive statistics is obtained from a single Cholesky factorisation of K. Table II summarises the model assessment results.

TABLE II: Model assessment metrics for the SE GP with MLE hyperparameters.

Metric	Value
Training RMSE	0.195
LOO RMSE	0.272
LOO log probability	−1.929

The discrepancy between training RMSE (0.195) and LOO RMSE (0.272) comes from the hyperparameters being tuned on all 15 observations; removing any point reveals slightly worse generalisation fit. The ratio of 1.39 indicates a mild but detectable optimism in the training fit. LOO log probability combines accuracy and calibration: a model with correct means but overconfident intervals is penalised, as is one with correct means but unnecessarily wide intervals. This makes the LOO log probability more informative than RMSE alone when assessing whether the posterior uncertainty is properly calibrated. In small-data applications, LOO cross-validation is preferable to marginal likelihood alone for the selection of a model, since marginal likelihood optimisation can itself overfit the hyperparameters when N is small.

D. Over-confidence in GP inference

A GP can produce smooth, plausible-looking predictions while being very over-confident. The kernel shape governs the posterior mean, while the posterior variance depends on how strongly the kernel assigns correlation between test and training points. These two quantities are coupled through ℓ, but not symmetrically.

With a large length scale, the kernel assigns high correlation between every test point and every training point. For $\ell = 15$ (three times the domain length), the kernel value $\exp\!\left(-d^2/(2\times 15^2)\right)$ is approximately 1 for any $d \le 5$. The kernel matrix K is approximately rank one, and the posterior covariance $\Sigma_{t^*} = k(X^*, X^*) - k(X^*, X)K^{-1}k(X, X^*)$ collapses down to almost zero everywhere, including outside the training range. Even though some regions contain no training data, the model still assigns negligible uncertainty to its extrapolations.

Fig. 7 compares the MLE model against the over-smoothed model. Both produce smooth posterior means within $[0, 5]$. The over-smoothed model's 95% CI is barely visible as a shaded band in the right panel, yet the true function exits the CI in both extrapolation regions. The model is confident, but it is wrong.

This failure is common in practice. The first reason is, hyperparameter selection by cross-validation on training error (rather than marginal likelihood) tends to prefer large ℓ; smooth models generalise well within the observed range and are rewarded accordingly. In addition, the marginal likelihood landscape penalises large ℓ only gradually; the right-hand side of Fig. 6 falls slowly, so moderate over-smoothing may not be strongly discouraged by the available data.

V. Bayesian Optimisation via Expected Improvement

A. Surrogate model and acquisition function

The SE GP from Section IV serves as a surrogate for the unknown function $g(x) = f(x)$. The objective here is to identify $x_{\max} = \arg\max_x g(x)$. The current best observed value is $\eta = \max_n t_n = 6.024$ at $x = 5.0$. The expected improvement acquisition function is:

$$\alpha_{\mathrm{EI}}(x) = \int_{-\infty}^{\infty} \max(t - \eta,\, 0)\, p(t|x, \mathcal{D})\, \mathrm{d}t. \tag{11}$$

Since $p(t|x, \mathcal{D}) = \mathcal{N}(t;\, \mu(x),\, \sigma^2(x))$, the integral has a closed form. Splitting at $t = \eta$ and substituting $z = (t - \mu)/\sigma$ gives [2]:

$$\alpha_{\mathrm{EI}}(x) = (\mu(x) - \eta)\, \Phi(Z) + \sigma(x)\, \phi(Z), \tag{12}$$

where $Z = (\mu(x) - \eta)/\sigma(x)$, Φ is the standard normal Cumulative Distribution Function (CDF), and ϕ is the standard normal Probability Density Function (PDF). The first term in (12) is the exploitation component, large when the posterior mean already exceeds η and the probability of improvement is high. The second term is the exploration component, large when posterior variance is high, even with a mean below η. When $\sigma(x) = 0$ the expression reduces to $\max(\mu(x) - \eta, 0)$.

The surrogate and acquisition function are shown in Fig. 8. The target $f(x) = 1.5x + \sin(x)$ is increasing ($f'(x) = 1.5 + \cos x \ge 0.5$ for all x), thus the maximum of the function situates beyond the right training boundary. The GP captures this as seen by the EI maintained at zero throughout $[0, 5]$, where the posterior mean lies below η and the surrogate is confident, at which point it rises (beyond $x = 5$) where the mean grows and variance increases. The EI attains a maximum

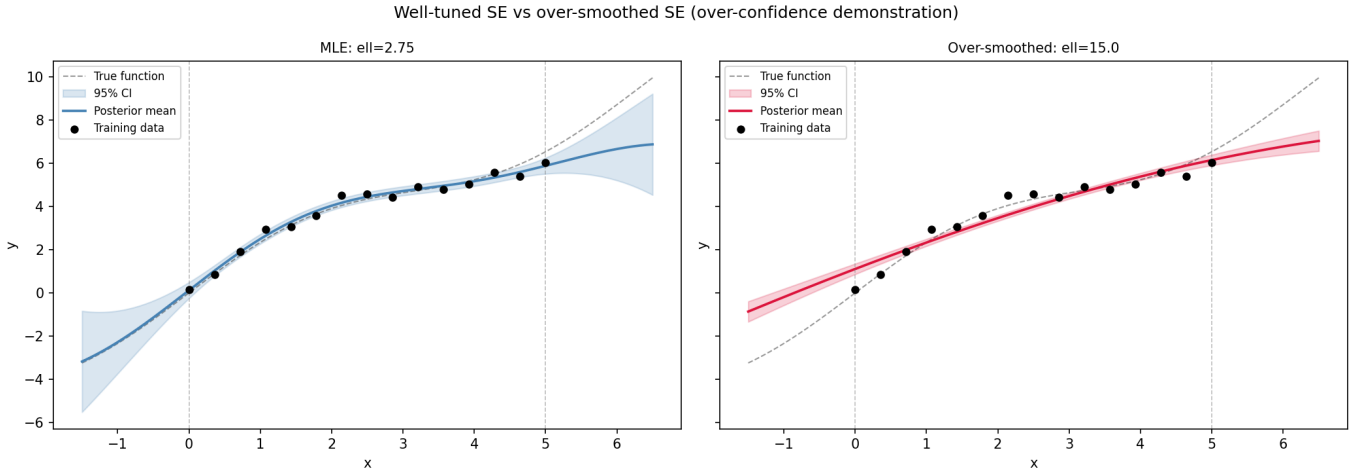


Fig. 7: MLE SE model ($\hat{\ell} = 2.753$, left) against an over-smoothed model ($\ell = 15$, right). Both produce smooth posterior means within the training range. The over-smoothed model's 95% CI is nearly invisible, while the true function (dashed) exits the CI in the extrapolation region.

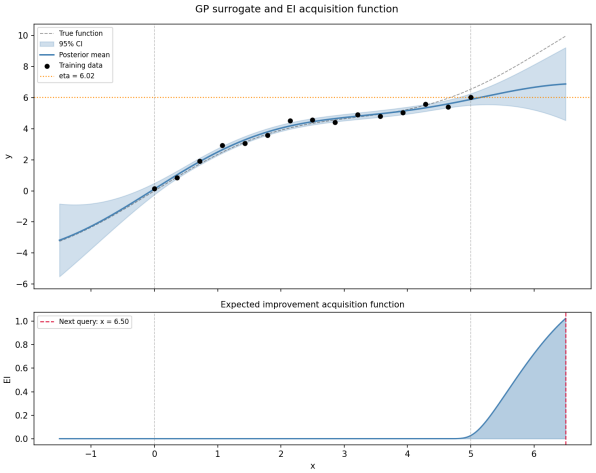


Fig. 8: Top: GP surrogate with 95% CI and current best value $\eta = 6.024$ (orange dotted line). Bottom: EI acquisition function. The EI peaks at $x = 6.5$ (red dashed line).

value of 1.024 at $x = 6.5$, the right boundary of the test domain.

B. Over-confident surrogate

The over-confident GP from Section IV ($\ell = 15$) brings forth a particular failure scenario. With posterior variance collapsed at ≈ 0 everywhere, the exploration term $\sigma(x)\phi(Z)$ in (12) vanishes. The EI reduces to $\max(\mu(x) - \eta, 0)$, a deterministic state depriving exploration.

Fig. 9 shows the result. The well-tuned model's EI (left panel) peaks in the extrapolation region and the exploration component is non-negligible throughout. The over-confident model's EI (right panel) is narrow everywhere. The algorithm would select the single point where the surrogate mean marginally exceeds η, then repeat that selection indefinitely.

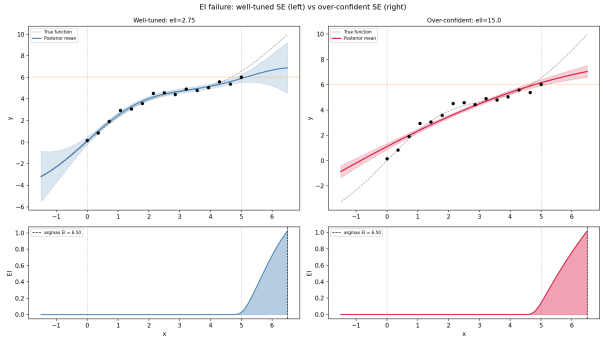


Fig. 9: EI acquisition functions for the well-tuned surrogate ($\hat{\ell} = 2.753$, left) and the over-confident surrogate ($\ell = 15$, right).

The over-confident model cannot distinguish between a region where the function is known to be low and a region it has never evaluated but confidently dismisses. If the true function had a peak inside the training range (in a gap between training inputs) the collapsed EI would assign a very small acquisition value there and perhaps never try discovering it.

C. Effect of noise level on optimisation

Noise standard deviation σ_d controls how confidently the GP interpolates through training inputs and in doing so, how the EI distributes its acquisition value between observed and unobserved regions. Fig. 10 compares three levels: $\sigma_d = 0.05$ (low), 0.232 (MLE), and 1.50 (high).

At $\sigma_d = 0.05$, the GP nearly interpolates the training data. Posterior variance within $[0, 5]$ is negligible, and the EI is effectively zero inside the training range. Every sampled input gets directed to the extrapolation region. The approach is efficient for a monotone increasing function but it would fail to locate an interior maximum that the initial grid did not sample well.

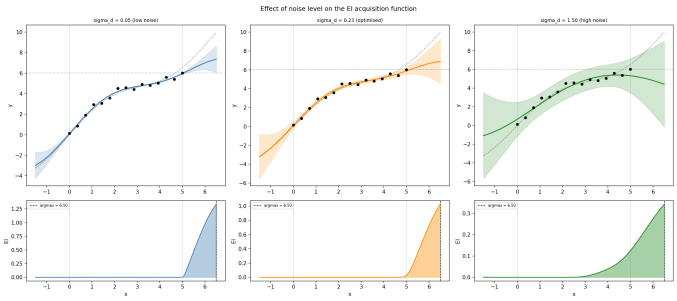


Fig. 10: GP posteriors (top row) and EI acquisition functions (bottom row) for three noise levels: $\sigma_d = 0.05$ (left), 0.232 (centre, MLE), and 1.50 (right). Low noise concentrates EI in the extrapolation region; high noise spreads acquisition broadly, driving exploration within the training range.

At $\sigma_d = 1.50$, the GP cannot constrain the function even at training locations. The EI is non-negligible from approximately $x = 4$ onwards and has a broad and flat profile, instead of a sharp peak. The algorithm re-evaluates training-range points because it is still uncertain about their values. This is exploration-dominated; the approach is flexible to observation noise but wastes evaluations in regions that were already visited.

The MLE value $\sigma_d = 0.232$ achieves a balance. The EI is concentrated in the right extrapolation region but retains sensitivity near the training boundary. The MLE noise estimate reflects the actual level of stochastic variation in the observations, and calibrating the surrogate to that level is what allows the acquisition function to allocate query budget most efficiently.

The exploration-exploitation balance depends on how well the surrogate's uncertainty is calibrated to the true observation noise. In practice, the noise level is rarely known a priori and must be inferred jointly with the kernel hyperparameters from the available data, making marginal-likelihood-based hyperparameter estimation quite an important component of any practical BO pipeline.

VI. Conclusion

This report examined the effect of kernel choice from prior construction through to Bayesian optimisation. Covariance matrices and sample paths provide qualitative evidence for kernel identification, but smoothness order above the first detectable level, exact hyperparameter values, and behaviour outside the observed domain remain unidentifiable from a finite sample set. On the regression benchmark $f(x) = 1.5x + \sin(x)$, the SE kernel achieves a log marginal likelihood of -11.32 against -21.11 for OU and -22.87 for the linear kernel; the ordering mirrors the degree of prior mismatch. Leave-one-out analysis gives a training-to-LOO RMSE ratio of 1.39, a detectable signal of hyperparameter overfit. A single poorly chosen length scale ($\ell = 15$ on a domain of length 5) suffices to collapse posterior variance globally, producing a 95% CI that excludes the true function in the extrapolation region while maintaining smooth predictions everywhere. Under expected improvement, the calibrated surrogate identifies $x = 6.5$ as the next query with $\alpha_{\text{EI}} = 1.024$; the over-confident surrogate's acquisition function is null everywhere, leading to no exploration being done. The noise level experiment confirms that the exploration-exploitation tradeoff is governed by both the acquisition function's formula and by the quality of the surrogate's uncertainty calibration.

References

[1] C. Rasmussen and C. Williams, "Gaussian Processes for Machine Learning," 2006. Available: https://gaussianprocess.org/gpml/chapters/RW.pdf.

[2] D. R. Jones, M. Schonlau, and W. J. Welch, "Efficient Global Optimization of Expensive Black-Box Functions," Journal of Global Optimization, vol. 13, no. 4, pp. 455–492, 1998, doi: https://doi.org/10.1023/a:1008306431147.

[3] E. Snelson and Z. Ghahramani, "Sparse Gaussian Processes using Pseudo-inputs," Advances in Neural Information Processing Systems, vol. 18, 2025, Available: https://papers.nips.cc/paper_files/paper/2005/hash/4491777b1aa8b5b32c2e8666dbe1a495-Abstract.html.

[4] A. Forrester and A. Keane, "Engineering Design via Surrogate Modelling A Practical Guide." Accessed: Apr. 27, 2026. [Online]. Available: https://download.e-bookshelf.de/download/0000/5794/92/L-G-0000579492-0015290213.pdf.